Computer Assignment 2 - Contact Mechanics

FEM-Solids, TME250, 2025

The task is to be carried out **individually** and a report, containing derivations, code (that you have written yourself) and conclusions, is to be handed in via Canvas as a **single pdf file** before **17:00 Tuesday, December 9, 2025**. In addition to including the source code in the pdf-file, we ask that you attach your code (e.g. Matlab files) in a zip-file. The reports, as well as the code, will automatically be sent to Ouriginal, to check for plagiarism. It is okay to discuss the implementation with other students, but **DO NOT** share code with each other.

A well written report, together with correct implementation, can give maximum 4 points. A late hand in, without prior agreement with the examiner, will give at maximum 3 points.

Contact us in good time if you need to postpone your hand in!

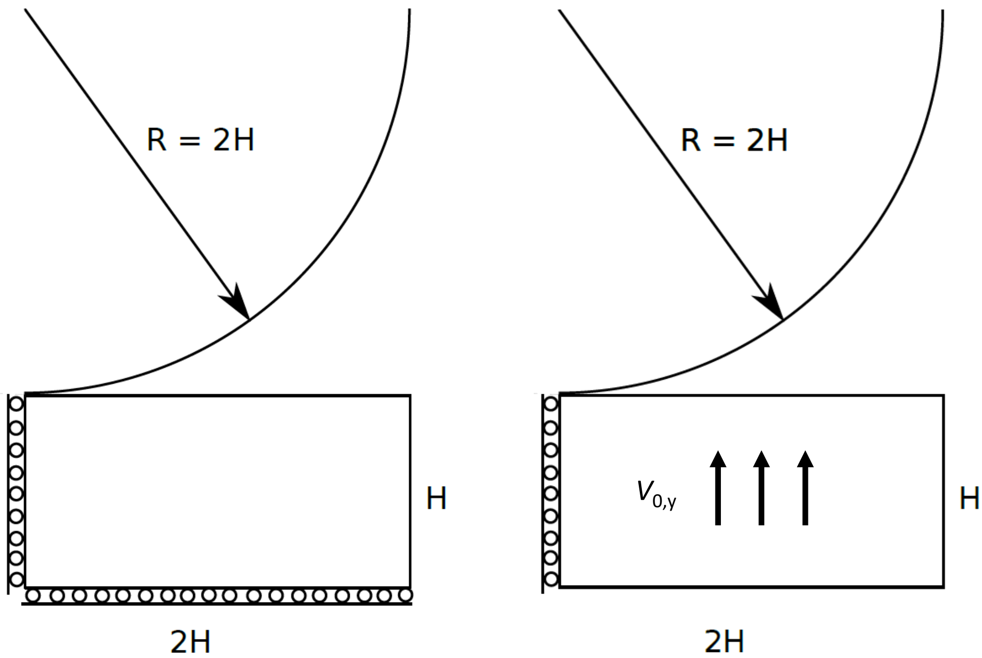

Figure 1: Geometry for the problem with the elastic box and rigid cylinder. In the static load scenario (left), the cylinder is pushed down onto the elastic block with boundary conditions according the figure. In the dynamic load scenario (right), the cylinder is only confined on its left boundary while given an initial vertical velocity $v_{0,\mathrm{y}}$ towards the now fixed cylinder. $H = 20\,\mathrm{mm}$ and $R = 40\,\mathrm{mm}$.

Problem formulation

Consider the two-dimensional domain in the figure[1], modeling a body under plane strain conditions. It is subjected to symmetric boundary conditions along the left and bottom edge.

A circular cylinder with radius 40 mm is pushed into the elastic block. First, the cylinder is placed with it's center at $\bar{\boldsymbol{x}} = (\bar{x}_1, \bar{x}_2) = (0, 60)$ mm. Origo is in the bottom left corner of the solid block, whereby the position of the cylinder corresponds to initial contact as shown in the picture.

The task is to study two different scenarios:

I) The cylinder is pushed down to the final position $\bar{\boldsymbol{x}} = (0, 55)$ mm, colliding with the

[1]Same domain as previous assignment.

rectangular block in the process. See Figure 1. The cylinder is rigid ($E_{\text{cyl}} = \infty$) and only the block is deformed elastically. Treat this case as quasi-static.

II) The ridgid cylinder remains in its initial position. The problem is treated dynamically and the elastic block has an initial velocity $\boldsymbol{v}_{\text{init}} = v_{0,\text{y}}\, \boldsymbol{e}_2$ that lets it move upwards. You can neglect gravitational effects.

The task is to study the problem using FEM and using the Lagrange multiplier method for the contact between the cylinder and the block. Implement the solution in a programming language of your choice. The Matlab toolbox CALFEM can be used. You can reuse the program *mesh.m* (available on the course home page) from the previous assignment that generates a uniform mesh.

1 Specific tasks

1.1 Scenario I

1. Implement and solve the problem in terms of incrementing the position of the cylinder from $\bar{\boldsymbol{x}} = (0, 60)$ mm to $\bar{\boldsymbol{x}} = (0, 55)$ mm. Use the Lagrange multiplier method for the contact using the linearized contact assumption. You may choose to compute the contact normal and the initial gap from the initial position of the finite element node ($\boldsymbol{u} = \boldsymbol{0}$), or from the previously converged increment ($\boldsymbol{u} = {}^{n-1}\boldsymbol{u}$). The latter will be more accurate.

2. Use the computed Lagrange multipliers to compute the *vertical* reaction force on the cylinder. Plot this total contact force vs. indentation as the cylinder is pressed into the elastic block.

3. Plot the von Mises stress distribution in the block and the displacement of the elastic block at the final step of the analysis (at maximum indentation).

1.2 Scenario II

1. Implement a code for running implicit dynamic simulations on the elastic block. In this first step, do not consider contact and instead constrain the vertical movement of the top edge of the block by a Dirichlet boundary condition. Show how the block deforms for a few selected steps. (You do not need to capture every step with this, make a choice of data to show.)

2. Compute the internal strain energy and the kinetic energy of the block and show them over time. What can you conclude for the energy balance of the block?

3. Add the contact formulation that you implemented for Scenario I to your dynamic code. Simulate Scenario II until the elastic block has bounced off the cylinder.

   (a) Show snapshots of the deformation. Your plots should include the grid in deformed shape and the cylinder. Make sure your plots represent physical dimensions correctly.

   (b) Compute the vertical reaction force acting on the cylinder (as done for the static load scenario) and show its evolution in time.

   (c) Also, plot the development of energy for the elastic body over time: internal, kinetic and total energy.

4. Finally, extend your code to explicit dynamics (selecting $\theta = 0$ in the generalized midpoint rule pertinent to the balance of momentum equation). Implement the more efficient algorithm incorporating a lumped mass matrix.

   Use the explicit code to run Scenario II and investigate the required time step. Show the same results as for the implicit simulation (snapshots of the displacements, history of the contact force, and development of internal, kinetic and total energy) for two the two cases of

(a) choosing a too large time-step, and
(b) choosing a sufficiently small time-step.

5. Comment on the results that you have obtained in this assignment. Consider the following points: What are the advantages of an implicit simulation, what are the advantages of an explicit simulation? What are ways to recognize that your time step is not sufficiently small? How do the mesh size and the time step interact in explicit simulations?

1.3 Parameters

The dynamic simulations can be very sensitive to the parameters used. We recommend starting with the set of parameters presented in Table 1 for the implicit dynamic simulation.

Parameter		Value	Unit
Young's modulus	E	50.0e3	MPa
Poisson's ratio	ν	0.3	-
density	ρ	8e−6	$\mathrm{kg\,mm^{-3}}$
final time	t_{final}	5e−3	s
time step	Δt	20e−6	s
initial vertical velocity	$v_{\mathrm{y},0}$	10e3	$\mathrm{mm\,s^{-1}}$
element edge length	l	2.0	mm

Table 1: Recommended parameters for the implicit dynamic simulations.

2 Hints

2.1 Contact constraints

For contact detection, consider a node located at $\boldsymbol{X}+\boldsymbol{u}$, where $\boldsymbol{X}$ is the original location and $\boldsymbol{u}$ the displacment. If the cylinder center is located at $\bar{\boldsymbol{x}}$, we may define the gap function for this node as follows (*Verify this by drawing a simple figure!*):

$$g(\boldsymbol{u}) = |(\boldsymbol{X}+\boldsymbol{u}) - \bar{\boldsymbol{x}}| - R. \tag{1}$$

Linearizing the gap function around the undeformed solution ($\boldsymbol{u}=0$), we obtain

$$g(\boldsymbol{u}) \approx g(\boldsymbol{0}) + \left.\frac{\partial g}{\partial \boldsymbol{u}}\right|_{\boldsymbol{u}=\boldsymbol{0}} \cdot = g_0 + \boldsymbol{N}_0 \cdot \boldsymbol{u}, \tag{2}$$

with the initial gap and the constant normal

$$g_0 = |\boldsymbol{X} - \bar{\boldsymbol{x}}| - R, \tag{3}$$

$$\boldsymbol{N}_0 = \frac{\boldsymbol{X}-\bar{\boldsymbol{x}}}{|\boldsymbol{X}-\bar{\boldsymbol{x}}|}, \tag{4}$$

Consider now the case that we seek to solve for the soltion $\boldsymbol{u}(t_n) = {}^{n}\boldsymbol{u}$ where $\bar{\boldsymbol{x}}(t_n) = {}^{n}\bar{\boldsymbol{x}}$. In order to improve accuracy, we may linearize the gap function around the previously converged solution (${}^{n-1}\boldsymbol{u} = \boldsymbol{u}(t_{n-1})$) as follows:

$$g(\boldsymbol{u}) \approx g({}^{n-1}\boldsymbol{u}) + \left.\frac{\partial g}{\partial \boldsymbol{u}}\right|_{\boldsymbol{u}={}^{n-1}\boldsymbol{u}} \cdot \left[\boldsymbol{u} - {}^{n-1}\boldsymbol{u}\right] = g_0 + \boldsymbol{N}_0 \cdot \boldsymbol{u}, \tag{5}$$

with the normal and initial gap

$$\boldsymbol{N}_0 = \frac{\boldsymbol{X} + {}^{n-1}\boldsymbol{u} - {}^{n}\bar{\boldsymbol{x}}}{|\boldsymbol{X} + {}^{n-1}\boldsymbol{u} - {}^{n}\bar{\boldsymbol{x}}|}, \tag{6}$$

$$g_0 = |\boldsymbol{X} + {}^{n-1}\boldsymbol{u} - {}^{n}\bar{\boldsymbol{x}}| - R - \boldsymbol{N}_0 \cdot {}^{n-1}\boldsymbol{u}, \tag{7}$$

respectively.

2.2 Consistent and lumped mass matrix

For the dynamic load scenario, the mass matrix for the discrete system is needed. The consistent (exactly integrated) element mass matrix for a CST-element becomes

$$\underline{M}^{\mathrm{e}} = \int_{\Omega_e} [\underline{N}^{\mathrm{e}}]^{\mathrm{T}} \rho \underline{N}^{\mathrm{e}} \, \mathrm{d}\Omega = \frac{A_e \rho}{12} \begin{bmatrix} 2 & 0 & 1 & 0 & 1 & 0 \\ 0 & 2 & 0 & 1 & 0 & 1 \\ 1 & 0 & 2 & 0 & 1 & 0 \\ 0 & 1 & 0 & 2 & 0 & 1 \\ 1 & 0 & 1 & 0 & 2 & 0 \\ 0 & 1 & 0 & 1 & 0 & 2 \end{bmatrix}, \tag{8}$$

where $\underline{N}^{\mathrm{e}}$ denotes the (12×2) matrix of linear basis functions, A_e is the area of the triangular element, and ρ is the material density. The global mass matrix is assembled completely analogously to that of the stiffness matrix.

In order to allow for an explicit time-integration, we adopt a lumped mass approximation for trivial inversion. The key ingredients is to construct a diagonal mass matrix that represents the exact same translational kinetic energy for a uniform velocity. For the CST element, it simply pertains to the approximation

$$\underline{M}^{\mathrm{e}} \approx \underline{M}^{\mathrm{e}}_{\mathrm{L}} = \frac{A_e \rho}{3} \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}, \tag{9}$$

2.3 Energy balance

Since there is no energy dissipation in the model setup, the system energy balance for the continuous (exact) solution is given as

$$\mathcal{E} + \mathcal{K} = \mathrm{const} \tag{10}$$

where $\mathcal{E}$ is the internal strain energy and $\mathcal{K}$ is the kinetic energy. These are respectively given by

$$\mathcal{E} = \frac{1}{2} \int_{\Omega} \boldsymbol{\varepsilon} : \boldsymbol{E} : \boldsymbol{\varepsilon} \, \mathrm{d}\Omega \,, \qquad \mathcal{K} = \frac{1}{2} \int_{\Omega} \rho \, \boldsymbol{v} \cdot \boldsymbol{v} \, \mathrm{d}\Omega \,. \tag{11}$$

In the FE-setting, it is easily verified that

$$\mathcal{E} = \frac{1}{2} \underline{a}^{\mathrm{T}} \, \underline{S} \, \underline{a} \,, \qquad \mathcal{K} = \frac{1}{2} \underline{v}^{\mathrm{T}} \, \underline{M} \, \underline{v} \,, \tag{12}$$

where $\underline{S}$ and $\underline{M}$ are the stiffness and mass matrices, respectively, and $\underline{a}$ and $\underline{v}$ are the nodal displacements and velocities.

2.4 The CFL-condition

In explicit analysis, numerical stability can be controlled in terms of satisfying the *Courant-Friedrichs–Lewy* (CFL) Condition. For the elastodynamic problem, it essentialy states that the time-step needs to be sufficiently small for a stress-wave not to pass through more than one element (anywhere in the domain) during one time-step. Hence, we require that

$$\Delta t \lessapprox \min \frac{h}{v}, \tag{13}$$

where h denotes the mesh-size and v denotes the wave-speed. For isotropic linear elasticty, the maximum local wave speed is that of a pressure wave $v = \sqrt{\frac{K}{\rho}}$, where K and ρ are the bulk modulus and density, respectively.